

Linear Algebra: Matrix Operations and Row Reduction

Lecture Notes

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Introduction to Matrix Operations

Matrices are fundamental objects in linear algebra. A matrix is a rectangular array of numbers arranged in rows and columns.

Definition: Matrix

An $m \times n$ matrix is a rectangular array of mn numbers arranged in m rows and n columns:

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}$$

where $a_{ij} \in \mathbb{R}$ is the entry in row i and column j . We denote this matrix as $A \in \mathbb{R}^{m \times n}$.

Matrix Addition and Scalar Multiplication

Definition: Matrix Addition

Let $A, B \in \mathbb{R}^{m \times n}$. The sum $A + B$ is the $m \times n$ matrix whose (i, j) -entry is $a_{ij} + b_{ij}$.

Definition: Scalar Multiplication

Let $A \in \mathbb{R}^{m \times n}$ and $c \in \mathbb{R}$. The scalar multiple cA is the $m \times n$ matrix whose (i, j) -entry is ca_{ij} .

Proposition: Properties of Matrix Operations

For matrices $A, B, C \in \mathbb{R}^{m \times n}$ and scalars $c, d \in \mathbb{R}$:

- (i) $A + B = B + A$ (commutativity)
- (ii) $(A + B) + C = A + (B + C)$ (associativity)
- (iii) $c(A + B) = cA + cB$ (distributivity)
- (iv) $(c + d)A = cA + dA$ (distributivity)
- (v) $c(dA) = (cd)A$ (associativity of scalars)

Matrix Multiplication

Definition: Matrix Multiplication

Let $A \in \mathbb{R}^{m \times n}$ and $B \in \mathbb{R}^{n \times p}$. The product AB is the $m \times p$ matrix whose (i, j) -entry is:

$$(AB)_{ij} = \sum_{k=1}^n a_{ik} b_{kj}$$

Remark: Matrix Multiplication is Not Commutative

In general, $AB \neq BA$. In fact, AB may be defined while BA is not, or both may be defined but produce different results.

Example: Simple Matrix Multiplication

Let $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$ and $B = \begin{pmatrix} 5 & 6 \\ 7 & 8 \end{pmatrix}$.

Then:

$$AB = \begin{pmatrix} 1 \cdot 5 + 2 \cdot 7 & 1 \cdot 6 + 2 \cdot 8 \\ 3 \cdot 5 + 4 \cdot 7 & 3 \cdot 6 + 4 \cdot 8 \end{pmatrix} = \begin{pmatrix} 19 & 22 \\ 43 & 50 \end{pmatrix}$$

Elementary Row Operations

Elementary row operations are the foundation of row reduction algorithms. These operations preserve the solution set of the system of linear equations represented by the matrix.

Definition: Elementary Row Operations

The three elementary row operations on a matrix are:

- (i) **Row Scaling (Type I):** Multiply a row by a nonzero scalar $c \in \mathbb{R}$, $c \neq 0$.
Notation: $R_i \rightarrow cR_i$.
- (ii) **Row Addition (Type II):** Add a multiple of one row to another row. Notation:
 $R_i \rightarrow R_i + cR_j$ where $i \neq j$ and $c \in \mathbb{R}$.
- (iii) **Row Swap (Type III):** Interchange two rows. Notation: $R_i \leftrightarrow R_j$ where $i \neq j$.

Theorem: Elementary Row Operations Preserve Solution Sets

If a matrix B is obtained from matrix A by a sequence of elementary row operations, then the systems of linear equations represented by A and B have the same solution set.

Example: Elementary Row Operations

Consider the augmented matrix:

$$\left(\begin{array}{cc|c} 2 & 4 & 6 \\ 1 & 3 & 5 \end{array} \right)$$

Applying $R_1 \leftrightarrow R_2$:

$$\left(\begin{array}{cc|c} 1 & 3 & 5 \\ 2 & 4 & 6 \end{array} \right)$$

Applying $R_2 \rightarrow R_2 - 2R_1$:

$$\left(\begin{array}{cc|c} 1 & 3 & 5 \\ 0 & -2 & -4 \end{array} \right)$$

Row Reduction and Row Echelon Form

Definition: Row Echelon Form (REF)

A matrix is in row echelon form if it satisfies:

- (i) All zero rows (rows with all entries equal to zero) appear at the bottom of the matrix.
- (ii) The first nonzero entry (called the pivot) in each nonzero row is to the right of the pivot in the row above it.
- (iii) All entries below a pivot are zero.

Definition: Reduced Row Echelon Form (RREF)

A matrix is in reduced row echelon form if it is in row echelon form and additionally satisfies:

- (i) Each pivot is equal to 1.
- (ii) All entries above and below each pivot are zero.

Remark: Uniqueness of RREF

Every matrix has a unique reduced row echelon form. In contrast, the row echelon form of a matrix is not unique.

Example: Row Echelon Form vs. RREF

The matrix

$$\begin{pmatrix} 1 & 2 & 3 \\ 0 & 4 & 5 \\ 0 & 0 & 6 \end{pmatrix}$$

is in row echelon form but not in reduced row echelon form.

The matrix

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

is in reduced row echelon form.

Gaussian Elimination

Gaussian elimination is an algorithm for transforming a matrix into row echelon form using elementary row operations.

Definition: Gaussian Elimination Algorithm

Given an $m \times n$ matrix A , Gaussian elimination proceeds as follows:

1. Identify the leftmost nonzero column. This is the first pivot column.
2. If the top entry in the pivot column is zero, swap rows to place a nonzero entry at the top.
3. Use row addition operations to make all entries below the pivot equal to zero.
4. Repeat the process for the remaining submatrix (ignoring the current row and all rows above it), moving to the right.
5. Continue until the matrix is in row echelon form.

Example: Gaussian Elimination

Consider the augmented matrix:

$$\left(\begin{array}{ccc|c} 2 & 1 & -1 & 8 \\ -3 & -1 & 2 & -11 \\ -2 & 1 & 2 & -3 \end{array} \right)$$

Step 1: Use R_1 as the pivot row. Eliminate below:

$$R_2 \rightarrow R_2 + \frac{3}{2}R_1$$

$$R_3 \rightarrow R_3 + R_1$$

$$\left(\begin{array}{ccc|c} 2 & 1 & -1 & 8 \\ 0 & 1/2 & 1/2 & 1 \\ 0 & 2 & 1 & 5 \end{array} \right)$$

Step 2: Use R_2 as the pivot row. Eliminate below:

$$R_3 \rightarrow R_3 - 4R_2$$

$$\left(\begin{array}{ccc|c} 2 & 1 & -1 & 8 \\ 0 & 1/2 & 1/2 & 1 \\ 0 & 0 & -1 & 1 \end{array} \right)$$

This matrix is now in row echelon form.

Proposition: Rank of a Matrix

The rank of a matrix A , denoted $\text{rank}(A)$, is the number of nonzero rows in any row echelon form of A . Equivalently, it is the number of pivot columns.

Forward Elimination

Forward elimination is the process of transforming a matrix into row echelon form. The algorithm works systematically from the top-left corner, moving right and down.

Theorem: Forward Elimination Termination

Every matrix can be reduced to row echelon form by Gaussian elimination in a finite number of steps.

Gauss-Jordan Elimination

Gauss-Jordan elimination is an extension of Gaussian elimination that transforms a matrix into reduced row echelon form.

Definition: Gauss-Jordan Elimination Algorithm

Gauss-Jordan elimination extends Gaussian elimination by:

1. First, apply Gaussian elimination to obtain row echelon form.
2. Then, perform back substitution using row operations to create zeros **above** each pivot as well.
3. Scale each pivot row so that the pivot equals 1.

Example: Gauss-Jordan Elimination

Continuing from the previous example, we have:

$$\left(\begin{array}{ccc|c} 2 & 1 & -1 & 8 \\ 0 & 1/2 & 1/2 & 1 \\ 0 & 0 & -1 & 1 \end{array} \right)$$

Step 1: Scale rows to have pivots equal to 1:

$$R_1 \rightarrow \frac{1}{2}R_1$$

$$R_2 \rightarrow 2R_2$$

$$R_3 \rightarrow -R_3$$

$$\left(\begin{array}{ccc|c} 1 & 1/2 & -1/2 & 4 \\ 0 & 1 & 1 & 2 \\ 0 & 0 & 1 & -1 \end{array} \right)$$

Step 2: Eliminate above the pivots, working from bottom-right to top-left:

$$R_2 \rightarrow R_2 - R_3$$

$$R_1 \rightarrow R_1 + \frac{1}{2}R_3$$

$$\left(\begin{array}{ccc|c} 1 & 1/2 & 0 & 7/2 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 1 & -1 \end{array} \right)$$

Step 3: Eliminate above remaining pivots:

$$R_1 \rightarrow R_1 - \frac{1}{2}R_2$$

$$\left(\begin{array}{ccc|c} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 1 & -1 \end{array} \right)$$

This matrix is now in reduced row echelon form. The solution to the system is $(x, y, z) = (2, 3, -1)$.

Back Substitution

Back substitution is the process of solving for the variables starting from the last equation and working backwards.

Definition: Back Substitution

Once a system is in row echelon form (or reduced row echelon form), back substitution involves:

1. Identifying free variables (columns without pivots).
2. Solving for the leading variables in terms of the free variables, starting from the bottom equation.

Example: Back Substitution with Free Variables

Consider the system in RREF:

$$\left(\begin{array}{ccccc|c} 1 & 0 & 2 & 0 & 3 & 5 \\ 0 & 1 & -1 & 0 & 4 & 2 \\ 0 & 0 & 0 & 1 & -2 & 3 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right)$$

The pivot columns are 1, 2, and 4. The free variables are x_3 and x_5 .

From the RREF form:

$$x_1 = 5 - 2x_3 - 3x_5$$

$$x_2 = 2 + x_3 - 4x_5$$

$$x_4 = 3 + 2x_5$$

Setting $x_3 = s$ and $x_5 = t$ (parameters), the general solution is:

$$\begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{pmatrix} = \begin{pmatrix} 5 \\ 2 \\ 0 \\ 3 \\ 0 \end{pmatrix} + s \begin{pmatrix} -2 \\ 1 \\ 1 \\ 0 \\ 0 \end{pmatrix} + t \begin{pmatrix} -3 \\ -4 \\ 0 \\ 2 \\ 1 \end{pmatrix}$$

Applications and Special Cases

Solving Linear Systems

Theorem: Solution Structure

For a system $A\mathbf{x} = \mathbf{b}$:

- (i) If $\text{rank}(A) < \text{rank}([A|\mathbf{b}])$, the system is inconsistent (no solution).
- (ii) If $\text{rank}(A) = \text{rank}([A|\mathbf{b}]) = n$ (number of columns), the system has a unique solution.
- (iii) If $\text{rank}(A) = \text{rank}([A|\mathbf{b}]) < n$, the system has infinitely many solutions.

Finding the Inverse of a Matrix

Definition: Matrix Inverse

An $n \times n$ matrix A is invertible if there exists an $n \times n$ matrix A^{-1} such that:

$$AA^{-1} = A^{-1}A = I_n$$

where I_n is the $n \times n$ identity matrix.

Theorem: Gauss-Jordan Method for Finding Inverses

To find A^{-1} for an invertible $n \times n$ matrix A :

1. Form the augmented matrix $[A|I_n]$.
2. Apply Gauss-Jordan elimination to transform this into $[I_n|A^{-1}]$.
3. The right half of the resulting matrix is A^{-1} .

Example: Finding a Matrix Inverse

Let $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$. We form:

$$\left[\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 3 & 4 & 0 & 1 \end{array} \right]$$

Apply $R_2 \rightarrow R_2 - 3R_1$:

$$\left[\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 0 & -2 & -3 & 1 \end{array} \right]$$

Apply $R_2 \rightarrow -\frac{1}{2}R_2$:

$$\left[\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 0 & 1 & 3/2 & -1/2 \end{array} \right]$$

Apply $R_1 \rightarrow R_1 - 2R_2$:

$$\left[\begin{array}{cc|cc} 1 & 0 & -2 & 1 \\ 0 & 1 & 3/2 & -1/2 \end{array} \right]$$

Therefore, $A^{-1} = \begin{pmatrix} -2 & 1 \\ 3/2 & -1/2 \end{pmatrix}$.

Determining Linear Dependence**Definition: Linear Independence and Dependence**

A set of vectors $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$ is linearly independent if the only solution to:

$$c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \dots + c_k\mathbf{v}_k = \mathbf{0}$$

is $c_1 = c_2 = \dots = c_k = 0$. Otherwise, the set is linearly dependent.

Proposition: Testing Linear Independence via Row Reduction

Arrange the vectors as columns of a matrix and perform row reduction. The vectors are linearly independent if and only if each column is a pivot column in the row echelon form.

Corollary: Rank and Nullity

For an $m \times n$ matrix A :

$$\text{rank}(A) + \text{nullity}(A) = n$$

where $\text{nullity}(A)$ is the dimension of the null space (kernel) of A .

Summary and Key Takeaways

Row reduction is the cornerstone of computational linear algebra. Through elementary row operations, we can:

1. Solve systems of linear equations.
2. Determine if a system is consistent or inconsistent.
3. Find the rank of a matrix and identify pivot and free variables.
4. Compute matrix inverses.
5. Test linear independence of vectors.
6. Find bases for fundamental subspaces.

Key Algorithms:

- **Gaussian Elimination:** Transforms a matrix into row echelon form (REF).
- **Gauss-Jordan Elimination:** Transforms a matrix into reduced row echelon form (RREF).
- **Back Substitution:** Extracts the solution from row echelon form.

The reduced row echelon form is unique for each matrix and provides complete information about the solution set of the corresponding linear system.